

NETWORK PROBLEM 1 :

IP ADDRESS CONFLICT (LAYER 3 NETWORK LAYER)

IP Address Conflict Issue occurs when two or more devices on a network try to use the same IP address. This situation can lead to communication disruptions between devices on the network, connection drops, and even complete unavailability of some devices.

Causes of IP Address Conflict:

1. Static IP Address Assignment:

- When a device in a network uses a static IP address, and another device manually assigns the same IP address, a conflict arises. ○ For example, if a computer in the network is assigned a static IP of 192.168.1.10, and another device tries to use the same IP, a conflict will occur.

2. Dynamic IP Address Assignment (DHCP):

- The **DHCP server** assigns dynamic IP addresses to devices. If the DHCP server gives the same IP to a device that was already manually assigned a static IP or reuses IP addresses incorrectly from its pool, a conflict can occur. ○ In most networks, DHCP automatically assigns IP addresses, but misconfigurations in the process can lead to IP conflicts.

3. DHCP Server Issues:

- If there are multiple DHCP servers on a network and they do not share the IP pool, conflicts can arise. For example, if one DHCP server assigns IPs from 192.168.1.100 to 192.168.1.200 while another DHCP server uses the same range, conflicts will occur.

4. New Devices Added to the Network:

- When a device's IP address is dynamically assigned, a previously used but now removed device's IP address could be reassigned to a new device, causing an IP conflict.

5. Hardware Failures in the Network:

- Hardware failures, particularly with the DHCP server, can also lead to IP conflicts. Similarly, misconfigurations in network cards or routers can cause conflicts.

Symptoms of IP Address Conflict:

1. Connection Drops:

- When a conflict occurs, devices on the network may fail to communicate properly, leading to dropped connections or very slow network speeds.

2. Error Messages from Network Devices:

- When a conflict is detected, some devices may display error messages. For instance, Windows devices might show error messages regarding network adapter IP conflicts.

3. Slow Performance:

- Networks with IP conflicts may experience slower performance as devices cannot send packets to each other properly, increasing network congestion.

4. Network Services Not Working:

- IP conflicts can cause network services to fail. For example, file sharing services or printers may become inaccessible due to the conflict.

Resolving IP Address Conflict:

1. Review Static IP Addresses:

- When assigning static IP addresses to devices in the network, make sure they do not overlap with the DHCP pool. For example, assign static IPs in the range from 192.168.1.1 to 192.168.1.10, and let the other devices automatically receive IP addresses via DHCP.
- Proper **IP Address Planning** helps prevent conflicts on the network.

2. DHCP Server Configuration:

- The DHCP server's IP pool should be correctly configured. The IP range assigned by the DHCP server should not overlap with the static IPs used in the network. Also, ensure that only one device manages the DHCP server.

3. Detecting IP Conflicts in the Network:

- Devices with IP conflicts should be identified by network administrators, and their IP addresses should be manually changed. This can be done by checking the **ARP cache** (Address Resolution Protocol table) or using **ping** commands across the network.

4. DHCP Lease Duration:

- The **DHCP lease duration** should be properly set. The lease duration determines how long a device can hold onto its assigned IP address. Shorter lease times can help release dynamic IPs more quickly and reduce conflict risks.

5. Monitor Network Routers and Browsers:

- By reviewing router log files, error messages related to conflicts can be detected. The router interface will show IP assignments and potential conflict details.

6. Update Network Hardware:

- Regular software updates for network hardware, especially routers and switches, are important. These updates ensure that the devices operate with the most secure and stable configurations.

7. Switch from Static IP to Dynamic IP:

- If IP conflicts occur frequently, switching from static IP usage to dynamic IPs (DHCP) can help. This ensures that devices automatically receive IP addresses from the DHCP server, reducing the chances of conflicts.

Simulation of problem: In this simulation, pc13 and pc14 were given the same ip address. In this way, an ip conflict was simulated. The orange circles on the ports also indicate this.

The screenshot displays the Cisco Packet Tracer interface. On the left, a network diagram shows a central switch connected to a router and two PCs (PC13 and PC14). Orange circles on the switch ports connected to the PCs indicate an IP conflict. On the right, the 'Event List' panel shows a table of network events. The table has columns for 'Vis.', 'Time(sec)', 'Last Device', 'At Device', and 'Type'. The following table represents the data shown in the Event List:

Vis.	Time(sec)	Last Device	At Device	Type
0.000	--	--	Switch5	DTP
0.001	--	Router7	Switch5	ARP
0.001	--	Switch5	Router7	DTP
0.001	--	--	PC13	ICMP
0.002	--	--	Switch5	CDP
0.002	--	--	Switch5	CDP
0.002	--	--	Switch5	CDP
0.003	--	Switch5	PC14	CDP
0.003	--	Switch5	PC13	CDP
0.003	--	Switch5	Router7	CDP
0.005	--	--	Switch5	DTP
0.005	--	--	Router7	CDP
0.005	--	--	PC13	ICMP
0.005	--	--	PC13	ICMP
0.006	--	Switch5	PC14	DTP
0.006	--	Router7	Switch5	CDP
1.007	--	--	PC13	ICMP

Below the event list, there are 'Play Controls' (Reset Simulation, Constant Delay, Play, Pause, Stop) and an 'Event List Filters - Visible Events' section. The bottom status bar shows the time as 02:30:31.674 and the simulation mode as Realtime.

Simulation of solution : Pc13 and Pc14 have different ip addresses (statically) on this simulation. In this scenario ,there is not ip conflict. We pinged pc14's ip address from pc13. It seems to work correctly in the command prompt.

The screenshot displays the Cisco Packet Tracer interface. The main workspace shows a network topology with a 2411 Router connected to a 2950-24TT Switch, which is then connected to two PCs, PC13 and PC14. A command prompt window is open on PC13, showing the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=9ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
```

The Event List panel on the right shows a series of events, including ARP requests and ICMP Echo (ping) events. The events are as follows:

Vis	Time(sec)	Last Device	At Device	Type
0.002	0.002	PC13	Switch4	ARP
0.002	0.002	Switch4	PC14	ARP
0.003	0.003	Switch4	Router6	ARP
0.003	0.003	Switch4	Router6	ARP
0.004	0.004	PC14	Switch4	ARP
0.005	0.005	Switch4	PC13	ARP
0.005	0.005	-	PC13	ICMP
0.006	0.006	PC13	Switch4	ICMP
0.007	0.007	Switch4	PC14	ICMP
0.008	0.008	PC14	Switch4	ICMP
0.009	0.009	Switch4	PC13	ICMP
1.010	1.010	-	PC13	ICMP
1.011	1.011	PC13	Switch4	ICMP
1.012	1.012	Switch4	PC14	ICMP
1.013	1.013	PC14	Switch4	ICMP
1.014	1.014	Switch4	PC13	ICMP